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Processing machine and slide mechanism

Abstract

A processing machine comprises: a cover that is provided with an opening and defines a processing area; a first door that is slidable between a first position in which the opening is opened and a second position in which the opening is closed; a second door that is disposed to overlap with the first door inside the processing area and slidable between the first position and a third position that is opposite to the first position with respect to the second position and in which the second door cooperates with the first door to close the opening; and a rail member that includes a grooved portion and an abutment that engage with the first door and is provided at the second door to allow the second door to slide with respect to the first door. The grooved portion and the abutment are disposed outside the processing area.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a processing machine and a slide mechanism.

BACKGROUND ART

(2) For example, Japanese Patent Application Laying-Open No. 2013-193197 (PTL 1) discloses a machine tool comprising an opening/closing door. The opening/closing door is supported by a guide rail and a door roller that slides on the guide rail to be laterally slidable.

CITATION LIST

Patent Literature

(3) [PTL 1] Japanese Patent Laying-Open No. 2013-193197

SUMMARY OF INVENTION

Technical Problem

(4) As disclosed in PTL 1 above, a known processing machine comprises a door slidably supported by a rail member. In such a processing machine, depending on the layout of the rail member, the rail member may be exposed to foreign matters such as chips or cutting oil as a workpiece is processed in a processing area. In that case, it is difficult to smoothly slide the door.

(5) In the machine tool disclosed in PTL 1, the guide rail is provided on a frame (or base member) of the machine that supports the opening/closing door slidably. In that case, problems may arise such as a need to ensure a space for providing the base member with an elongate guide rail.

(6) Accordingly, it is an object of the present invention to solve the above problem and provide a processing machine allowing a door to smoothly slide. Another object of the present invention is to provide a slide mechanism that can solve problems caused by providing a rail member to a base member that supports a movable body slidably.

Solution to Problem

(7) According to the present invention, a processing machine comprises: a cover body that is provided with an opening and defines a processing area; a first door that is slidable in a predetermined direction between a first position in which the opening is opened and a second position in which the opening is closed; a second door that is disposed to overlap with the first door inside the processing area and slidable in the predetermined direction between the first position in which the opening is opened and a third position that is opposite to the first position with respect to the second position and in which the second door cooperates with the first door to close the opening; and a rail member that includes an engagement unit that engages with the first door and is provided at the second door to allow the second door to slide with respect to the first door in the predetermined direction. The engagement unit is disposed outside the processing area.

(8) According to the thus configured processing machine, the engagement unit of the rail member that engages with the first door is disposed outside the processing area and can thus be prevented from being exposed to foreign matters such as chips or cutting oil as a workpiece is processed in the processing area. Thus, the engagement unit is kept clean, and the second door can smoothly be slid with respect to the first door.

(9) Preferably, the second door includes an outer surface that faces the first door when the first door and the second door are in the first position. The rail member is provided at the second door without projecting from the outer surface toward the first door.

(10) According to the thus configured processing machine, the first door and the second door can be provided closer to each other, and the processing area can be enhanced in sealability.

(11) Preferably, the engagement unit includes an open end that is located opposite to the first door when the first door is in the second position and the second door is in the third position and that opens in one direction along the predetermined direction. The second door further includes a door body portion provided with the rail member, and a lid member that is attached to the door body portion and closes an opening defined by the open end.

(12) According to the thus configured processing machine, the first door can be engaged with the

engagement unit through the open end when the processing machine is assembled. Further, the lid member that is attached to the door body portion and closes the opening defined by the open end can prevent foreign matters such as chips or cutting oil from entering the engagement unit.
(13) Preferably, the lid member includes a labyrinth portion that extends toward the first door and forms a labyrinth structure with the first door.
(14) According to the thus configured processing machine, a number of parts can be reduced by providing the lid member that closes the opening defined by the open end with a labyrinth portion for providing a labyrinth structure integrally.
(15) Preferably, the predetermined direction is a horizontal direction. The opening includes a front opening located in front, with respect to the machine, of the processing area, and an upper opening contiguous to the front opening and located above the processing area. The second door includes a front door portion and an upper door portion that close the front opening and the upper opening, respectively, when the second door is in the third position. Of the front and upper door portions, the front door portion is provided with the rail member.
(16) According to the thus configured processing machine, the front door portion closes the front opening located in front, with respect to the machine, of the processing area, and foreign matters such as chips or cutting oil are likely to scatter toward the front door portion. Providing the rail member at the front door portion of the front and upper door portions can more effectively prevent the rail member from being exposed to foreign matters such as chips or cutting oil.
(17) According to the present invention, a slide mechanism comprises: a base member; a movable body that is slidable in a predetermined direction; and a rail member that is provided at the movable body and engages with the base member so that the movable body is slidable in the predetermined direction with respect to the base member.
(18) According to the thus configured slide mechanism, providing the rail member to the movable body can eliminate the necessity of providing the rail member to the base member. This can eliminate problems caused by providing the rail member to the base member.
Advantageous Effects of Invention
(19) Thus, according to the present invention, a processing machine allowing a door to smoothly slide can be provided. Further, according to the present invention, a slide mechanism that can solve problems caused by providing a rail member to a base member that supports a movable body slidably, can be provided.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view of a processing machine according to an embodiment of the present invention.
(2) FIG. **2** is a perspective view of a first door and a second door in an open state in the processing machine shown in FIG. **1**.
(3) FIG. **3** is a perspective view of the first door and the second door in a closed state in the processing machine shown in FIG. **1**.
(4) FIG. **4** is a cross section of the processing machine taken along and seen in a direction indicated by an arrow of a line IV-IV indicated in FIG. **3**.
(5) FIG. **5** is a perspective view of a traveling wheel provided at the first door shown in FIG. **4** and an attachment structure for the traveling wheel.
(6) FIG. **6** is another perspective view of the traveling wheel provided at the first door shown in FIG. **4** and the attachment structure for the traveling wheel.
(7) FIG. **7** is a cross section of the processing machine taken along and seen in a direction indicated by an arrow of a line VII-VII indicated in FIG. **3**.

(8) FIG. **8** is an exploded view of assembling the second door shown in FIG. **7**.
(9) FIG. **9** is a cross section of the processing machine taken along and seen in a direction indicated by an arrow of a line IX-IX in FIG. **3**.
(10) FIG. **10** is a cross section of the processing machine taken along and seen in a direction indicated by an arrow of a line X-X indicated in FIG. **3**.
(11) FIG. **11** is a cross section of the processing machine taken along and seen in a direction indicated by an arrow of a line XI-XI indicated in FIG. **3**.

DESCRIPTION OF EMBODIMENTS

(12) An embodiment of the present invention will now be described with reference to the drawings. In the drawings referenced below, identical or equivalent members are identically denoted.
(13) FIG. **1** is a perspective view of a processing machine according to an embodiment of the present invention. FIG. **2** is a perspective view of a first door and a second door in an open state in the processing machine shown in FIG. **1**. FIG. **3** is a perspective view of the first door and the second door in a closed state in the processing machine shown in FIG. **1**.
(14) Referring to FIGS. **1** to **3**, a processing machine **100** is a numerically control (NC) processing machine in which a variety of types of operations for processing a workpiece are automated through numerical control by a computer. Processing machine **100** is a composite processing machine having a turning function using a fixed tool and a milling function using a rotating tool.
(15) In the present specification, an axis parallel to a rightward/leftward direction (or widthwise direction) of processing machine **100** and extending in a horizontal direction is referred to as a “Z axis,” an axis parallel to a frontward/rearward direction (or depthwise direction) of processing machine **100** and extending in a horizontal direction is referred to as a “Y axis,” and an axis extending in a vertical direction is referred to as an “X axis.” In FIG. **1**, a rightward direction is referred to as a “+Z-axis direction” and a leftward direction is referred to as a “−Z-axis direction.” In FIG. **1**, a frontward direction with respect to the sheet of the figure is referred to as a “+Y-axis direction” and a rearward direction with respect thereto is referred to as a “−Y-axis direction.” In FIG. **1**, an upward direction is referred to as a “+X-axis direction,” and a downward direction is referred to as a “−X-axis direction.” The X-axis, the Y-axis, and the Z-axis are three axes orthogonal to one another.
(16) Processing machine **100** includes a cover body **21**. Cover body **21** defines a processing area **110** and configures an external appearance of processing machine **100**. Processing area **110** is a space in which a workpiece is processed and which is generally sealed so as to prevent foreign matters such as chips or cutting oil from leaking outside processing area **110** as the workpiece is processed.
(17) As shown in FIG. **2**, cover body **21** is provided with an opening **50**. Opening **50** defines an opening allowing processing area **110** to have an interior in communication with outside of processing area **110**. An operator can access the interior of processing area **110** from outside processing area **110** through opening **50**.
(18) Opening **50** has a front opening **51** and an upper opening **52**. Front opening **51** is located in front, with respect to the machine, of processing area **110** (or on a side of processing area **110** in the +Y-axis direction). Upper opening **52** is located above processing area **110** (or on a side thereof in the +X-axis direction). Upper opening **52** is contiguous to front opening **51**.
(19) Cover body **21** includes a first door **31**, a second door **41**, a side cover **27**, and a magazine cover **26**.
(20) First door **31** and second door **41** are disposed in opening **50**. First door **31** is slidable in a predetermined direction between a first position J in which opening **50** is opened and a second position K in which opening **50** is closed. Second door **41** is disposed to overlap with first door **31** inside processing area **110**. Second door **41** is slidable in the predetermined direction between first position J in which opening **50** is opened and a third position L that is opposite to first position J with respect to second position K and in which second door **41** cooperates with first door **31** to

close opening **50**.

(21) First door **31** and second door **41** are slidable independently of each other. First door **31** and second door **41** are slidable in the same direction. First door **31** and second door **41** are slidable in a horizontal direction. First door **31** and second door **41** are slidable in the Z-axis direction.

(22) First door **31** and second door **41** may be manually slid doors or automatic doors slid by a cylinder or a motor or a similar actuator.

(23) As shown in FIGS. **1** and **3**, when first door **31** is in second position K and second door **41** is in third position L, opening **50** is closed. First door **31** is disposed at a position adjacent to second door **41** in the +Z-axis direction. First door **31** and second door **41** define and form processing area **110** with an intra-machine cover disposed in processing area **110** on right and left sides with respect to the machine, a telescopic cover disposed on a rear side in processing area **110** and deformable as a tool spindle (not shown) moves in the X- and Z-axis directions, a ceiling cover disposed above processing area **110**, and the like.

(24) First door **31** includes a front door portion **32** and an upper door portion **33**. When first door **31** is in second position K, front door portion **32** closes front opening **51** shown in FIG. **2**, and upper door portion **33** closes upper opening **52** shown in FIG. **2**. Front door portion **32** is generally in the form of a plate parallel to the X-Z plane. Front door portion **32** is provided with a transparent window **34** (see FIG. **1**). Upper door portion **33** is generally in the form of a plate parallel to the Y-Z plane. An upper end portion of front door portion **32** and a front end portion (an end portion in the +Y-axis direction) of upper door portion **33** are contiguous to form a corner portion.

(25) Second door **41** includes a front door portion **42** and an upper door portion **43**. When second door **41** is in third position L, front door portion **42** closes front opening **51** shown in FIG. **2**, and upper door portion **43** closes upper opening **52** shown in FIG. **2**. Front door portion **42** is generally in the form of a plate parallel to the X-Z plane. Front door portion **42** is provided with a transparent window **44** (see FIG. **1**). Upper door portion **43** is generally in the form of a plate parallel to the Y-Z plane. An upper end portion of front door portion **42** and a front end portion (an end portion in the +Y-axis direction) of upper door portion **43** are contiguous to form a corner portion.

(26) First door **31** in second position K and second door **41** in third position L overlap with each other in a first range **310** in the Z-axis direction.

(27) More specifically, an end portion of second door **41** in the +Z-axis direction overlaps with an end portion of first door **31** in the −Z-axis direction inside processing area **110**. An end portion of front door portion **42** of second door **41** in the +Z-axis direction and an end portion of front door portion **32** of first door **31** in the −Z-axis direction are opposite to each other in the Y-axis direction. An end portion of upper door portion **43** of second door **41** in the +Z-axis direction and an end portion of upper door portion **33** of first door **31** in the −Z-axis direction are opposite to each other in the X-axis direction.

(28) As shown in FIGS. **1** to **3**, side cover **27** and magazine cover **26** are provided on opposite sides, respectively, of opening **50** in the Z-axis direction. Side cover **27** is provided at a position adjacent in the +Z-axis direction to first door **31** in second position K. Inside side cover **27** is accommodated a spindle for rotating a workpiece, or a tail stock for supporting the center of rotation of the workpiece, etc. Magazine cover **26** is provided at a position adjacent in the −Z-axis direction to second door **41** in third position L. Inside magazine cover **26** are accommodated a magazine for storing a plurality of tools used to process a workpiece etc.

(29) As shown in FIG. **2**, when first door **31** and second door **41** are in first position J, opening **50** is in an open state. First door **31** and second door **41** are disposed inside side cover **27**. Processing area **110** is opened to an outside of processing area **110** through opening **50** in the open state.

(30) First door **31** in first position J and second door **41** in first position J overlap with each other in a second range **320** in the Z-axis direction. Second range **320** in the Z-axis direction is larger in length than first range **310** in the Z-axis direction shown in FIG. **3**. The end portion of first door **31** in the −Z-axis direction is exposed from side cover **27**. An end portion of second door **41** in the −Z-

axis direction is exposed from first door **31**.
(31) FIG. **4** is a cross section of the processing machine taken along and seen in a direction indicated by an arrow of a line Iv-Iv indicated in FIG. **3**. Referring to FIGS. **3** and **4**, second door **41** has an outer surface **46** and an inner surface **47**. When second door **41** is in third position L, outer surface **46** is located outside processing area **110**. Inner surface **47** is located inside processing area **110** rearwardly of outer surface **46**.
(32) Inner surface **47** faces the −Y-axis direction, and outer surface **46** faces the +Y-axis direction. When first door **31** and second door **41** are in first position J, outer surface **46** faces first door **31** in second range **320** shown in FIG. **2**. When first door **31** is in second position K and second door **41** is in third position L, outer surface **46** faces first door **31** in first range **310** shown in FIG. **3**.
(33) Second door **41** includes a door body portion **45**. When second door **41** is in second position K, door body portion **45** serves as a main portion of second door **41** closing opening **50**. Door body portion **45** is composed of a front door portion **42** and an upper door portion **43**.
(34) Processing machine **100** further comprises a rail member **61**. Rail member **61** is provided at second door **41**. Rail member **61** is provided at door body portion **45**. Rail member **61** is provided at front door portion **42**. Of front door portion **42** and upper door portion **43**, front door portion **42** is provided with rail member **61**.
(35) Rail member **61** has an elongate shape having a length in the Z-axis direction. Rail member **61** is provided at a position closer in the X-axis direction to the lower end of front door portion **42** than the upper end of front door portion **42**.
(36) Rail member **61** is provided at second door **41** without projecting from outer surface **46** toward first door **31**. An internal space **115** is formed between inner surface **47** and outer surface **46**. Rail member **61** is accommodated in internal space **115**. Second door **41** has a notch **49** that opens to outer surface **46**. Internal space **115** and processing area **110** are isolated from each other with inner surface **47** interposed therebetween. Internal space **115** is in communication with an outside of processing area **110** through notch **49**.
(37) Processing machine **100** comprises a plurality of rail members **61** (**61**A, **61**B). A rail member **61**A and a rail member **61**B are disposed in internal space **115** and opposite to each other in a vertical direction. Rail member **61**B is provided above rail member **61**A. Rail member **61**A is provided immediately below an opening formed by notch **49**. Rail member **61**B is provided immediately above the opening formed by notch **49**.
(38) Rail member **61**A is formed of a plate bent along a plurality of straight lines extending in the Z-axis direction. Rail member **61**B is composed of a block.
(39) Rail member **61**A has a grooved portion **62**. Grooved portion **62** forms a groove extending in the Z-axis direction. Grooved portion **62** is open in the +X-axis direction (or upward). Rail member **61**B has an abutment **63**. Abutment **63** is in the form of a plane extending in the Z-axis direction. Abutment **63** is composed of a plane facing the −X-axis direction (or downward) and parallel to the Y-Z plane.
(40) Grooved portion **62** and abutment **63** are disposed outside processing area **110**. Grooved portion **62** and abutment **63** are disposed in internal space **115**. Grooved portion **62** and abutment **63** are isolated from processing area **110** with inner surface **47** interposed therebetween. Grooved portion **62** and abutment **63** correspond to an “engagement unit” in the present invention.
(41) FIGS. **5** and **6** are perspective views of a traveling wheel provided at the first door shown in FIG. **4** and an attachment structure for the traveling wheel. Referring to FIGS. **4** to **6**, processing machine **100** further comprises a first traveling wheel **64** (**64**A, **64**B) and a second traveling wheel **65**.
(42) First and second traveling wheels **64** and **65** are provided at first door **31**. First and second traveling wheels **64** and **65** are provided at front door portion **32**.
(43) First and second traveling wheels **64** and **65** are provided at positions closer to the lower end of front door portion **32** than the upper end of front door portion **32** in the X-axis direction. First

and second traveling wheels **64** and **65** are provided in the Z-axis direction at positions closer to the end portion of first door **31** in the −Z-axis direction than an end portion of first door **31** in the +Z-axis direction. As shown in FIG. **3**, when first door **31** is in second position K and second door **41** is in third position L, first and second traveling wheels **64** and **65** are located in first range **310**. First and second traveling wheels **64** and **65** are each formed for example of a cam follower.
(44) First traveling wheel **64** is provided rotatably about an axis of rotation **121** extending in the X-axis direction. First traveling wheels **64**A and **64**B are spaced from each other in the Z-axis direction.
(45) Second traveling wheel **65** is provided rotatably about an axis of rotation **122** extending in the Y-axis direction. Second traveling wheel **65** is provided above first traveling wheel **64**. Second traveling wheel **65** is provided between first traveling wheels **64**A and **64**B in the Z-axis direction.
(46) Grooved portion **62** and abutment **63** engage with first door **31**. Rail member **61** allows second door **41** to slide with respect to first door **31**.
(47) More specifically, first traveling wheel **64** is fitted in grooved portion **62**. First traveling wheels **64**A and **64**B are fitted in grooved portion **62** at positions spaced from each other in the Z-axis direction. Fitting first traveling wheel **64** into grooved portion **62** restricts a positional relationship between first door **31** and second door **41** in the Y-axis direction. Second traveling wheel **65** is disposed under abutment **63**. Second traveling wheel **65** abuts on abutment **63**. Causing second traveling wheel **65** to abut on abutment **63** restricts a positional relationship between first door **31** and second door **41** in the X-axis direction.
(48) Second door **41** is supported slidably in the Z-axis direction by first traveling wheel **64** (**64**A, **64**B) moving along grooved portion **62** while rotating about axis of rotation **121**, and second traveling wheel **65** moving along abutment **63** while rotating about axis of rotation **122**.
(49) Processing machine **100** further comprises a first block **66** and a second block **69**. First and second traveling wheels **64** and **65** are attached to first door **31** (or front door portion **32**) via first block **66** and second block **69**.
(50) First block **66** is attached to front door portion **32**. First block **66** is opposite in the Y-axis direction to the opening formed by notch **49**. Second block **69** is attached to first block **66**. Second block **69** projects from first block **66** in the −Y-axis direction and enters internal space **115** through notch **49**. First and second traveling wheels **64** and **65** are attached to second block **69**.
(51) Processing machine **100** further comprises a frame **67** and a bolt **68**. Frame **67** is attached to front door portion **32**. Frame **67** is provided below first block **66**. Bolt **68** extends in the X-axis direction. An end portion of bolt **68** in the +X-axis direction is secured to first block **66**. An end portion of bolt **68** in the −X-axis direction bolts frame **67**.
(52) This configuration can change a length of bolt **68** projecting from frame **67**, and hence a position (or level) of first block **66** in the X-axis direction. Thus, first and second traveling wheels **64** and **65** can be positionally adjusted in the vertical direction.
(53) According to this configuration, rail member **61** has grooved portion **62** and abutment **63** outside processing area **110**. Therefore, as shown in FIG. **3**, when first door **31** is in second position K and second door **41** is in third position L, and a workpiece is processed in processing area **110**, grooved portion **62** and abutment **63** can be prevented from being exposed to foreign matters such as chips or cutting oil as the workpiece is processed. This allows first and second traveling wheels **64** and **65** to smoothly rotate along grooved portion **62** and abutment **63**, respectively, and second door **41** to slide more reliably. In the present embodiment, in particular, rail member **61** configured to have grooved portion **62** and abutment **63** disposed outside processing area **110** is adopted at front door portion **42** at which foreign matters such as chips or cutting oil easily scatter. This more effectively allows second door **41** to slide more reliably.
(54) Further, rail member **61** is provided without projecting from outer surface **46** of second door **41** toward first door **31**. This configuration allows first door **31** and second door **41** to be provided closer to each other, and can enhance processing area **110** in sealability when first door **31** is in

second position K and second door **41** is in third position L. This can more reliably prevent processing area **110** from leaking foreign matters such as chips or cutting oil.

(55) Rail member **61** is provided at second door **41**, which is a movable body slidable in the Z-axis direction. This configuration can reduce limitation in designing first door **31** as it is unnecessary to provide rail member **61** that is an elongate body to first door **31** that is a base member that supports second door **41** slidably.

(56) FIG. **7** is a cross section of the processing machine taken along and seen in a direction indicated by an arrow of a line VII-VII indicated in FIG. **3**. FIG. **8** is an exploded view of assembling the second door shown in FIG. **7**.

(57) Referring to FIGS. **7** and **8**, door body portion **45** (or front door portion **42**) further includes a side surface **48**. Side surface **48** faces the +Z-axis direction. An end of side surface **48** in the −Y-axis direction is contiguous to inner surface **47**, and an end of side surface **48** in the +Y-axis direction is contiguous to outer surface **46**.

(58) Grooved portion **62** and abutment **63** have an open end **70**. When first door **31** is in second position K and second door **41** is in third position L, open end **70** is located at an end of first range **310** shown in FIG. **3**. Open end **70** is opposite to first door **31** in the Y-axis direction. Open end **70** is open in one direction along the Z axis. Open end **70** is open in the +Z-axis direction. Open end **70** defines an opening in side surface **48**.

(59) Second door **41** further includes a lid member **71**. Lid member **71** is attached to door body portion **45**. Lid member **71** is detachably attachable to door body portion **45** with a bolt or the like. Lid member **71** is attached to front door portion **42**. Lid member **71** is provided so as to close the opening defined by open end **70**.

(60) Lid member **71** includes a lid body portion **72** and a labyrinth portion **73**. Lid body portion **72** is in the form of a plate parallel to the X-Y plane. Lid body portion **72** is brought to overlap with side surface **48** to close the opening defined by open end **70**.

(61) Labyrinth portion **73** extends from lid body portion **72** toward first door **31** and has the extension with a tip bent and thus projecting in the −Z-axis direction. Labyrinth portion **73** has a bent structure in which a plate material parallel to the X-Y plane and a plate material parallel to the X-Z plane form a corner.

(62) First door **31** has an inner surface **37**. Inner surface **37** faces outer surface **46** of second door **41**. First door **31** includes a labyrinth member **76**. Labyrinth member **76** extends from inner surface **37** toward second door **41** (or outer surface **46**) and has the extension with a tip bent and thus projecting in the +Z-axis direction. When first door **31** is in second position K and second door **41** is in third position L, labyrinth portion **73** and labyrinth member **76** are opposite to each other with a gap therebetween in the Z-axis direction and the Y-axis direction to configure a labyrinth structure.

(63) According to this configuration, when processing machine **100** (cover body **21**) is assembled, first and second traveling wheels **64** and **65** can be inserted to grooved portion **62** and abutment **63**, respectively, through open end **70**. Further, attaching lid member **71** to lid body portion **72** and closing by lid body portion **72** the opening defined by open end **70** can prevent foreign matters such as chips or cutting oil from entering grooved portion **62** from processing area **110**.

(64) Further, labyrinth portion **73** forming a labyrinth structure with labyrinth member **76** of first door **31** is provided to lid member **71** in one piece. This allows processing machine **100** to be composed of a reduced number of parts and hence manufactured at a reduced cost.

(65) FIG. **9** is a cross section of the processing machine taken along and seen in a direction indicated by an arrow of a line IX-IX indicated in FIG. **3**. Referring to FIGS. **3** and **9**, processing machine **100** further comprises a support member **80**, a lower rail **81**, and a plurality of traveling wheels **82**, **83**, **84**.

(66) Support member **80** is provided under first door **31** and second door **41**. Support member **80** is fixed to an oil pan (not shown) in processing area **110**. Lower rail **81** is supported by support

member **80**. Lower rail **81** extends in the Z-axis direction. Lower rail **81** extends in the Z-axis direction across first position J, second position K, and third position L.
(67) Traveling wheel **82** is provided at second door **41**. Traveling wheel **82** is provided at front door portion **42**. Traveling wheel **82** is provided below rail member **61**. Traveling wheel **82** is provided at a position closer in the Z-axis direction to the end portion of second door **41** in the −Z-axis direction than the end portion of second door **41** in the +Z-axis direction.
(68) Traveling wheels **83** and **84** are provided at first door **31**. Traveling wheels **83** and **84** are provided at front door portion **32**. Traveling wheels **83** and **84** are provided below first and second traveling wheels **64** and **65**. Traveling wheels **83** and **84** are provided apart from each other in the Z-axis direction. Traveling wheel **83** is provided at a position closer in the Z-axis direction to the end portion of first door **31** in the −Z-axis direction than the end portion of first door **31** in the +Z-axis direction. Traveling wheel **84** is provided at a position away from traveling wheel **83** in the +Z-axis direction. Traveling wheel **84** is provided at a position closer in the Z-axis direction to the end portion of first door **31** in the +Z-axis direction than the end portion of first door **31** in the −Z-axis direction.
(69) Traveling wheels **82**, **83**, **84** are provided rotatably about an axis of rotation **123** parallel to the Y-axis direction. For example, traveling wheel **82**, **83**, **84** is a pulley formed to have a groove recessed radially inwards towards axis of rotation **123** and also extending circumferentially about axis of rotation **123**.
(70) Traveling wheels **82**, **83**, **84** are disposed on lower rail **81**. Second door **41** is supported slidably in the Z-axis direction by traveling wheel **82** moving along lower rail **81** while rotating about axis of rotation **123**. First door **31** is supported slidably in the Z-axis direction by traveling wheels **83** and **84** moving along lower rail **81** while rotating about axis of rotation **123**.
(71) First door **31** and second door **41** are supported from below by a single rail or lower rail **81**. Traveling wheels **82**, **83**, and **84** are aligned in the Z-axis direction. Wherever first door **31** and second door **41** may be positioned, traveling wheel **82** is located at a position closer to the −Z-axis direction than traveling wheels **83** and **84** are.
(72) FIG. **10** is a cross section of the processing machine taken along and seen in a direction indicated by an arrow of a line X-X indicated in FIG. **3**. Referring to FIGS. **3** and **10**, processing machine **100** further comprises a rail member **86** and a traveling wheel **88**.
(73) Rail member **86** is provided at first door **31**. Rail member **86** is provided at upper door portion **33**. Rail member **86** has an elongate shape having a length in the Z-axis direction. Rail member **86** has a grooved portion **87**. Grooved portion **87** forms a groove that opens in the +Y-axis direction and extends in the Z-axis direction.
(74) Traveling wheel **88** is provided at second door **41**. Traveling wheel **88** is provided at upper door portion **43**. Traveling wheel **88** is provided at a position closer in the Z-axis direction to the end portion of second door **41** in the +Z-axis direction than the end portion of second door **41** in the −Z-axis direction. When first door **31** is in second position K and second door **41** is in third position L, traveling wheel **88** is located in first range **310** shown in FIG. **3**.
(75) Traveling wheel **88** is provided rotatably about an axis of rotation **124** extending in the Y-axis direction. Traveling wheel **88** is composed for example of a cam follower. Rail member **86** receives the weight of second door **41** via traveling wheel **88**.
(76) Grooved portion **87** engages with second door **41**. More specifically, traveling wheel **88** is fitted in grooved portion **87**. Second door **41** is supported slidably in the Z-axis direction by traveling wheel **88** moving along rail member **86** while rotating about axis of rotation **124**.
(77) According to this configuration, rail member **86** provided at first door **31** receives the weight of second door **41** via traveling wheel **88**. This can prevent the end portion of second door **41** in the +Z-axis direction from tilting downward when first door **31** shown in FIG. **3** is in second position K and second door **41** shown in FIG. **3** is in third position L.
(78) FIG. **11** is a cross section of the processing machine taken along and seen in a direction

indicated by an arrow of a line XI-XI indicated in FIG. **3**. Referring to FIGS. **3** and **11**, processing machine **100** comprises a support member **91**, a ceiling rail **92**, a plurality of traveling wheels **94** (**94**A, **94**B), a ceiling rail **96**, and a plurality of traveling wheels **98** (**98**A, **98**B).
(79) Support member **91** is provided at a position adjacent to upper door portions **33** and **43** in the −Y-axis direction. Support member **91** is fixed to a frame member (not shown) disposed on the ceiling of processing machine **100**. Ceiling rails **92** and **96** are provided on support member **91**.
(80) Ceiling rail **92** is opposite to upper door portion **43** in the Y-axis direction. Ceiling rail **96** is opposite to upper door portion **33** in the Y-axis direction. Ceiling rail **92** is provided below ceiling rail **96**. Ceiling rails **92** and **96** each have an elongate shape having a length in the Z-axis direction. Ceiling rail **92** is provided in the Z-axis direction across first position J, second position K, and third position L. Ceiling rail **96** is provided in the Z-axis direction across first position J and second position K.
(81) Ceiling rail **92** has a plurality of grooved portions **93** (**93**A, **93**B). Grooved portion **93** forms a groove extending in the Z-axis direction. Grooved portion **93** is disposed outside processing area **110**. Grooved portion **93**A forms a groove that opens in the −X-axis direction and extends in the Z-axis direction. Grooved portion **93**B forms a groove that opens in the +Y-axis direction and extends in the Z-axis direction. Grooved portion **93**A is provided above grooved portion **93**B.
(82) Ceiling rail **96** has a plurality of grooved portions **97** (**97**A, **97**B). Grooved portion **97** forms a groove extending in the Z-axis direction. Grooved portion **97** is disposed outside processing area **110**. Grooved portion **97**A forms a groove that opens in the −X-axis direction and extends in the Z-axis direction. Grooved portion **97**B forms a groove that opens in the +Y-axis direction and extends in the Z-axis direction. Grooved portion **97**A is provided above grooved portion **97**B.
(83) Traveling wheel **94** is provided at second door **41**. Traveling wheel **94** is provided at upper door portion **43**. Traveling wheel **94** is composed for example of a cam follower. Traveling wheel **94**A is provided rotatably about an axis of rotation **125** extending in the X-axis direction. Traveling wheel **94**B is provided rotatably about an axis of rotation **126** extending in the Y-axis direction. Traveling wheels **94**A and **94**B are paired and provided at two locations apart from each other in the Z-axis direction.
(84) Traveling wheel **94** is fitted in grooved portion **93**. Traveling wheel **94**A is fitted in grooved portion **93**A, and traveling wheel **94**B is fitted in grooved portion **93**B. Second door **41** is supported slidably in the Z-axis direction by traveling wheel **94**A moving along grooved portion **93**A while rotating about axis of rotation **125** and traveling wheel **94**B moving along grooved portion **93**B while rotating about axis of rotation **126**.
(85) Traveling wheel **98** is provided at first door **31**. Traveling wheel **98** is provided at upper door portion **33**. Traveling wheel **98** is composed for example of a cam follower. Traveling wheel **98**A is provided rotatably about an axis of rotation **127** extending in the X-axis direction. Traveling wheel **98**B is provided rotatably about an axis of rotation **128** extending in the Y-axis direction. Traveling wheels **98**A and **98**B are paired and provided at two locations apart from each other in the Z-axis direction.
(86) Traveling wheel **98** is fitted in grooved portion **97**. Traveling wheel **98**A is fitted in grooved portion **97**A, and traveling wheel **98**B is fitted in grooved portion **97**B. First door **31** is supported slidably in the Z-axis direction by traveling wheel **98**A moving along grooved portion **97**A while rotating about axis of rotation **127**, and traveling wheel **98**B moving along grooved portion **97**B while rotating about axis of rotation **128**.
(87) The structure of processing machine **100** and the structure of the slide mechanism applied to processing machine **100** described above will now be summarized below. According to the present embodiment, processing machine **100** comprises: cover body **21** that is provided with opening **50** and defines processing area **110**; first door **31** that is slidable in a horizontal direction as a predetermined direction between first position J in which opening **50** is opened and second position K in which opening **50** is closed; second door **41** that is disposed to overlap with first door **31**

inside processing area **110** and slidable in the horizontal direction between first position J in which opening **50** is opened and third position L that is opposite to first position J with respect to second position K and in which second door **41** cooperates with first door **31** to close opening **50**; and rail member **61** that includes grooved portion **62** and abutment **63** as an engagement unit that engages with first door **31**, and is provided at second door **41** to allow second door **41** to slide with respect to first door **31** in the horizontal direction. Grooved portion **62** and abutment **63** are disposed outside processing area **110**.
(88) Further, according to the present embodiment, a slide mechanism applied to processing machine **100** comprises first door **31** as a base member, second door **41** as a movable body slidable in a horizontal direction as a predetermined direction, and rail member **61** that is provided at second door **41** and engages with first door **31** so that second door **41** is slidable with respect to first door **31** in the horizontal direction.
(89) Although first door **31** and second door **41** have been described in the present embodiment as being slidable in a horizontal direction, this is not a limitation, and the first door and the second door may be slidable for example in a vertical direction.
(90) Further, the processing machine of the present invention is not limited to a composite processing machine, and may for example be a lathe, a vertical machining center or a horizontal machining center, or may be an AM/SM hybrid processing machine capable of additively and subtractively manufacturing a workpiece.
(91) Further, the slide mechanism of the present invention is not limited to a door of a processing machine and is applicable for example to a door of an automobile or an automatic door.
(92) It should be understood that the embodiments disclosed herein are illustrative and non-restrictive in any respect. The scope of the present invention is defined by the terms of the claims, rather than the above description, and is intended to include any modifications within the meaning and scope equivalent to the terms of the claims.
INDUSTRIAL APPLICABILITY
(93) The present invention is applied to a processing machine comprising an openable and closable door or a slide mechanism such as a door.
REFERENCE SIGNS LIST
(94) **21** cover body, **26** magazine cover, **27** side cover, **31** first door, **32**, **42** front door portion, **33**, **43** upper door portion, **34**, **44** transparent window, **37**, **47** inner surface, **41** second door, **45** door body portion, **46** outer surface, **48** side surface, **49** notch, **50** opening, **51** front opening, **52** upper opening, **61**, **61**A, **61**B, **86** rail member, **62**, **87**, **93**, **93**A, **93**B, **97**, **97**A, **97**B grooved portion, **63** abutment **64**, **64**A, **64**B first traveling wheel, **65** second traveling wheel, **66** first block, **67** frame, **68** bolt, **69** second block, **70** open end, **71** lid member, **72** lid body portion, **73** labyrinth portion, **76** labyrinth member **80**, **91** support member, **81** lower rail, **82**, **83**, **84**, **88**, **94**, **94**A, **94**B, **98**, **98**A, **98**B traveling wheel, **92**, **96** ceiling rail, **100** processing machine, **110** processing area, **115** internal space, **121**, **122**, **123**, **124**, **125**, **126**, **127**, **128** axis of rotation, **310** first range, **320** second range, J first position, K second position, L third position.

Claims

1. A processing machine comprising: a cover body that is provided with an opening and defines a processing area in which a workpiece is processed; a first door that is slidable in a predetermined direction between a first position in which the opening is opened and a second position in which the opening is closed; a second door that is disposed to overlap with the first door inside the first door with respect to the processing area and slidable in the predetermined direction between the first position in which the opening is opened and a third position that is opposite to the first position with respect to the second position and in which third position the second door cooperates with the first door to close the opening; and a rail member that includes an engagement unit that

engages with the first door so as to guide the sliding of the first door, which rail member is provided at the second door to allow the second door to be slidable with respect to the first door in the predetermined direction, the engagement unit being disposed outside the processing area, the second door including an outer surface and an inner surface separated by an internal space formed therebetween, the second door also being provided with a notch opening to the outer surface, the outer surface being located outside the processing area and facing the first door when the first door and the second door are in the first position, the inner surface being located inside the processing area, the rail member being accommodated in the internal space, and the first door engaging with the engagement unit in the internal space via the notch.

2. The processing machine according to claim 1, wherein the rail member is provided at the second door without projecting from the outer surface toward the first door.

3. The processing machine according to claim 1, wherein the engagement unit includes an open end that is located opposite to the first door when the first door is in the second position and the second door is in the third position and that opens in one direction along the predetermined direction, and the second door further includes a door body portion provided with the rail member, and the second door includes a lid member that is attached to the door body portion and that closes an opening defined by the open end.

4. The processing machine according to claim 3, wherein the lid member includes a labyrinth portion that extends toward the first door and forms a labyrinth structure with the first door.

5. The processing machine according to claim 1, wherein the predetermined direction is a horizontal direction, the opening includes a front opening located in front, with respect to the machine, of the processing area, and an upper opening contiguous to the front opening and located above the processing area, the second door includes a front door portion and an upper door portion that close the front opening and the upper opening, respectively, when the second door is in the third position, and of the front and upper door portions, the front door portion is provided with the rail member.

6. A processing machine comprising: a cover body that is provided with an opening and defines a processing area in which a workpiece is processed; a first door that is slidable in a predetermined direction between a first position in which the opening is opened and a second position in which the opening is closed; a second door that is disposed to overlap with the first door inside the first door with respect to the processing area and slidable in the predetermined direction between the first position in which the opening is opened and a third position that is opposite to the first position with respect to the second position and in which third position the second door cooperates with the first door to close the opening; and a rail member that includes an engagement unit that engages with the first door so as to guide the sliding of the first door, which rail member is provided at the second door to allow the second door to be slidable with respect to the first door in the predetermined direction, the engagement unit being disposed outside the processing area, wherein the engagement unit includes an open end that is located opposite to the first door when the first door is in the second position and the second door is in the third position and that opens in one direction along the predetermined direction, and wherein the second door further includes a door body portion provided with the rail member, and the second door includes a lid member that is attached to the door body portion and that closes an opening defined by the open end.

7. The processing machine according to claim 6, wherein the second door includes an outer surface that faces the first door when the first door and the second door are in the first position, and the rail member is provided at the second door without projecting from the outer surface toward the first door.

8. The processing machine according to claim 6, wherein the lid member includes a labyrinth portion that extends toward the first door and forms a labyrinth structure with the first door.

9. The processing machine according to claim 6, wherein the predetermined direction is a horizontal direction, the opening includes a front opening located in front, with respect to the

machine, of the processing area, and an upper opening contiguous to the front opening and located above the processing area, the second door includes a front door portion and an upper door portion that close the front opening and the upper opening, respectively, when the second door is in the third position, and of the front and upper door portions, the front door portion is provided with the rail member.